

Operant Conditioning

Learning and Behavior (PSY/NSC 3320)

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Key Vocab

Operant Conditioning

“A type of learning in which the future frequency (or probability) of a behavior is affected by its consequences”

Key Vocab

Operant Behaviors

“behaviors that are influenced by their consequences”

Operant Conditioning

“effects of a behavior’s consequences upon the behavior”

Key Vocab

Operant Behaviors

Flipping a light switch (influenced by whether a light turns on after)

Operant Conditioning

Not flipping a switch anymore because the light never turns on

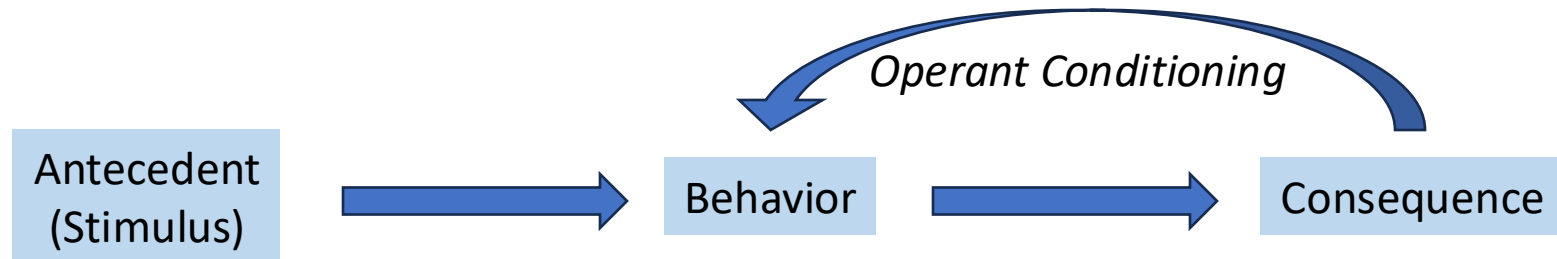
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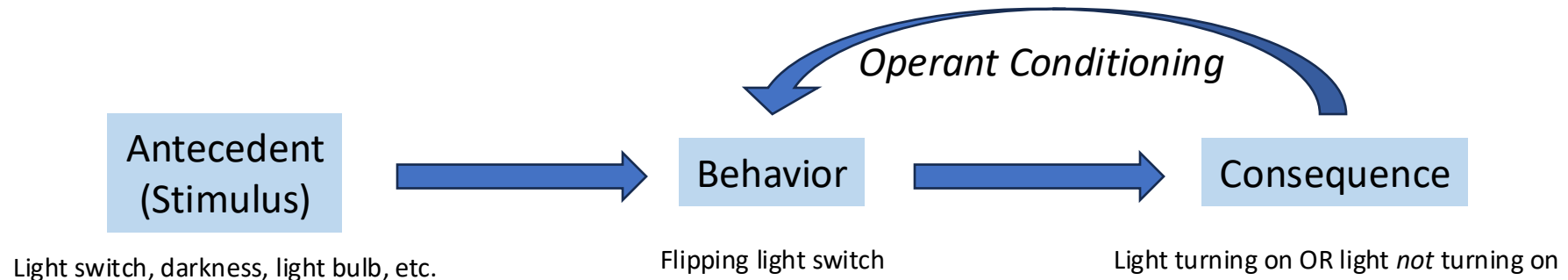
Key Vocab

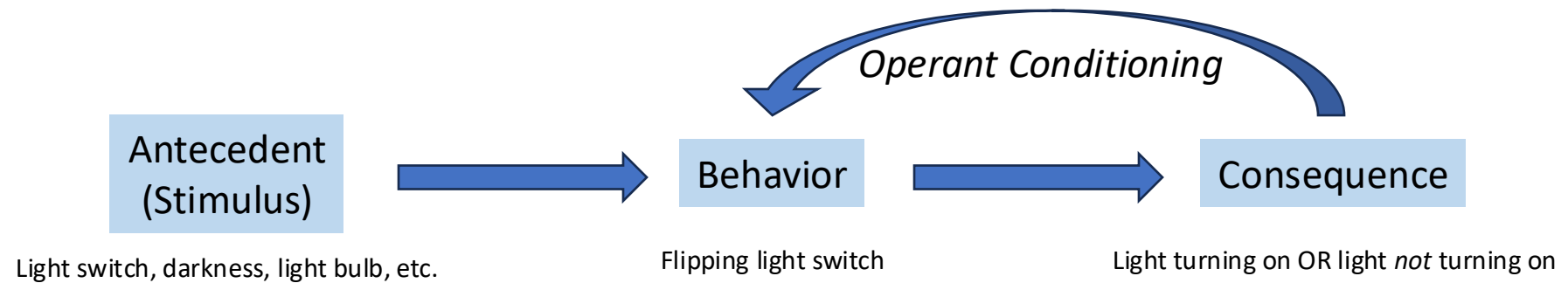
Operant Behaviors

Flipping a light switch (influenced by whether a light turns on after)

Operant Conditioning

Not flipping a switch anymore because the light never turns on





Key Vocab

Operant Behaviors

“Emitted” by organism

Voluntary responses to antecedent

Hear phone ring, answer it

Vs

Respondent Behaviors

“Elicited” by stimulus

Involuntary responses to a stimulus

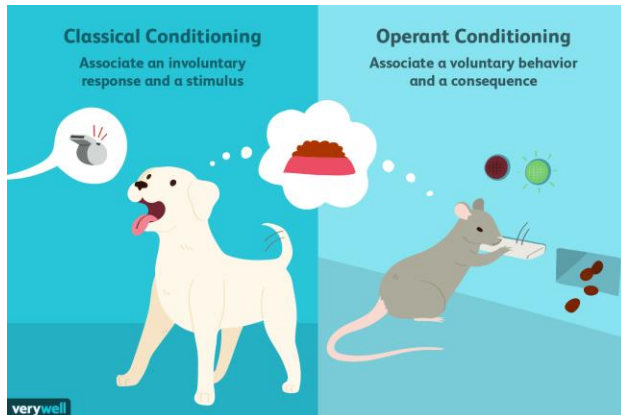
Salivate

Key Concepts

Respondent (Classical) Conditioning

Pairing two stimuli to each elicit a response

Bell:Food → Salivate



Operant Conditioning

Pairing a voluntary response to a consequence

Lever → Food

Respondent Conditioning

Result: Bell → Salivation



Operant Conditioning

Result: Telephone Ring → Telephone Answered (by 3 rings)



Key Concepts

Thorndike's Law of Effect

“behaviors leading to a satisfying state of affairs are strengthened, while behaviors leading to an unsatisfying or annoying state of affairs are weakened”

First 'operant conditioning'-related concept
Preceded operant conditioning and led to it

Key Concepts

Thorndike's Law of Effect

“behaviors leading to a satisfying state of affairs are strengthened, while behaviors leading to an unsatisfying or annoying state of affairs are weakened”

Thorndike's Puzzle Box Experiment

Place food outside box with gate attached to pedal

Cat learns to press foot pedal to open gate

Behavior → satisfying state of affairs

So, behavior strengthened

Time to escape decreased across many trials



Key Concepts

B.F. Skinner

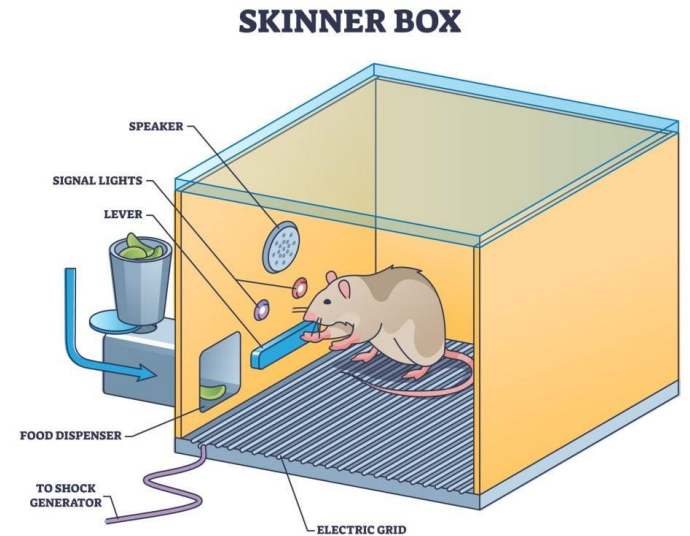
Operant conditioning boxes
Improved on Thorndike's Puzzle Box

"Free operant" procedure

Animal 'free' to decide when to emit behavior (lever-pressing)

Experimenter can manipulate consequences, etc.

Study animals via experimenter-controlled behavior $\leftrightarrow$ consequence paradigms (as Thorndike somewhat did) instead of via reflexes (as most did at the time)



Practical Aspects

Types of Consequences

- Reinforcers (S^R)
 - Follows a behavior
 - Future frequency of that behavior **increases**
- Punishers (S^P)
 - Follows a behavior
 - Future frequency of that behavior **decreases**

Lever Press (R) → Food Pellet (S^R)

Practical Aspects

- Reinforcers (S^R)
 - Follows a behavior
 - Future frequency of that behavior increases
- Punishers (S^P)
 - Follows a behavior
 - Future frequency of that behavior decreases

Reinforcer = consequence
Reinforcement = type of procedure
Same for punisher/punishment

Lever Press (R) $\rightarrow$ Food Pellet (S^R)

A rat presses a lever more because it learns that the lever produces food

A baby cries less because the parents always ignore it when it does so

A dog's mother ignores it when it cries, so the dog cries more as a result



Which of these are reinforcers and which are punishers?

Jeff gives his turtle, Terry, a banana (Terry hates bananas) each time it comes out to greet him, which makes the turtle do so less often

The likelihood of the squirrel eating the dog's food decreases because the dog chases it away every time

Reinforcer

A rat **presses a lever more** because it learns that the lever produces food

Punisher

A baby **cries less** because the parents always ignore it when it does so

Reinforcer

A dog's mother ignores it when it cries, so the dog **cries more** as a result



Which of these are
reinforcers and which are
punishers?

Punisher

The **likelihood** of the squirrel eating the dog's food **decreases** because the dog chases it away every time

Punisher

Jeff gives his turtle, Terry, a banana each time it comes out to greet him, which makes the turtle **do so less often** (Terry does not like bananas)

Practical Aspects

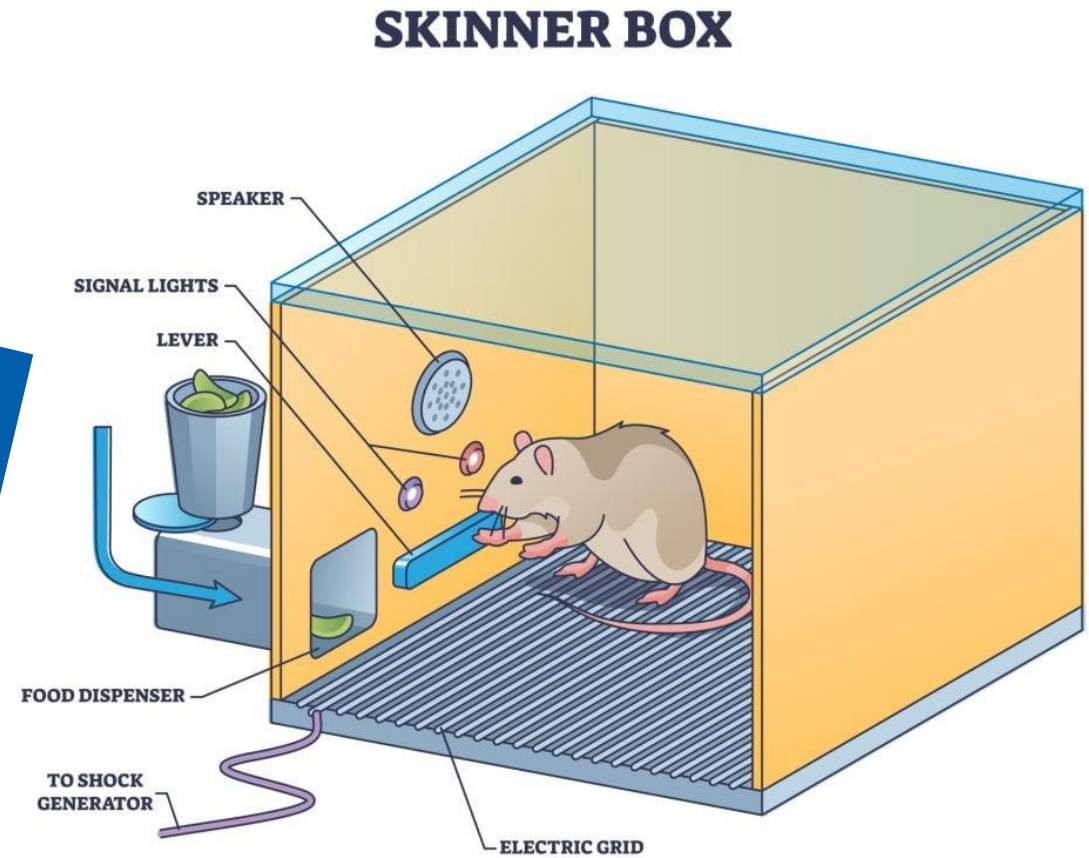
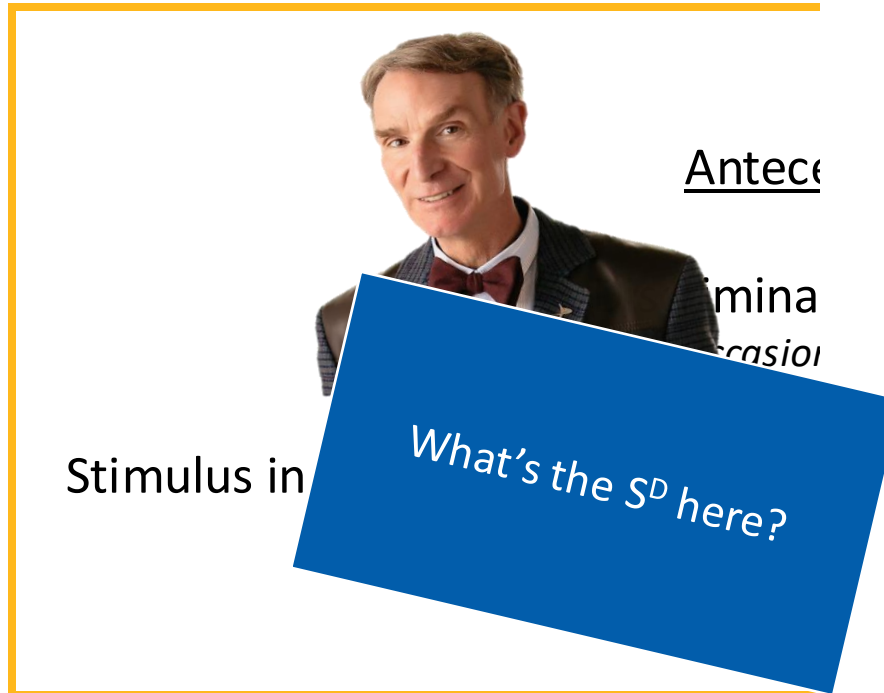
Antecedents

AKA – Discriminative Stimulus (S^D)

'Occasion Setters'

Stimulus in the presence of which responses are reinforced and in the absence of which they are not reinforced

Practical Aspects



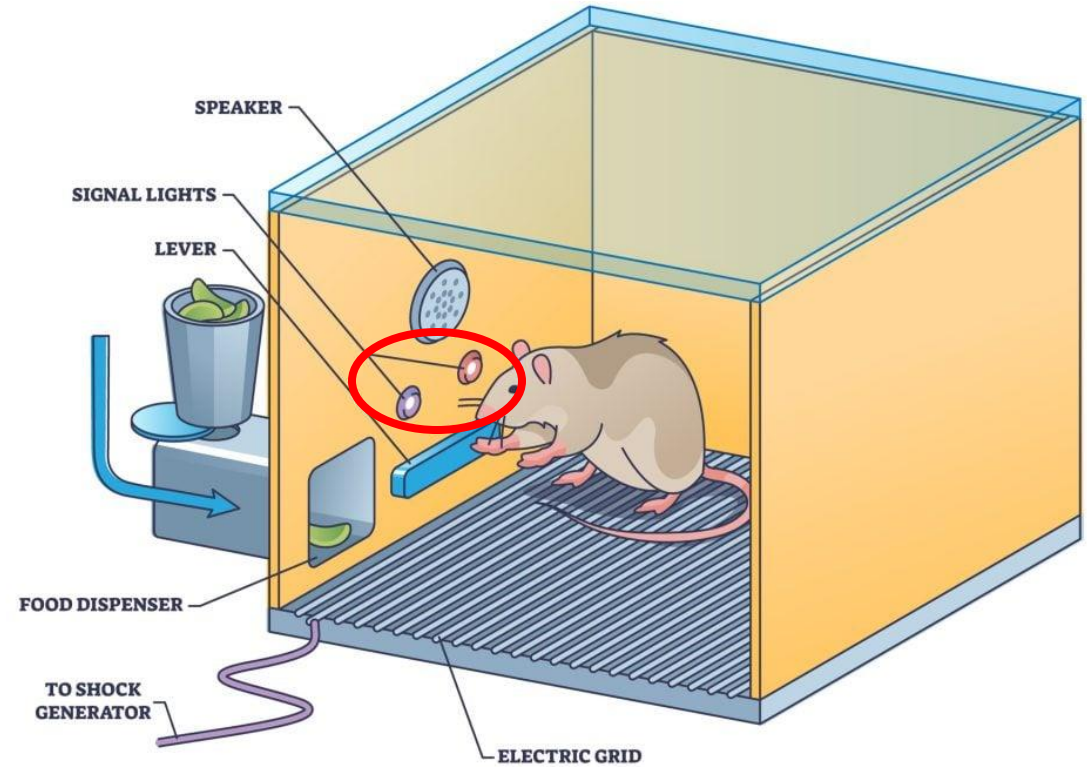
Practical Aspects

Antecedent

AKA – Discriminative Stimulus
'Occasion'

Stimulus in the presence of which response is reinforced
absence of which the response is not reinforced

SKINNER BOX



Practical Aspects

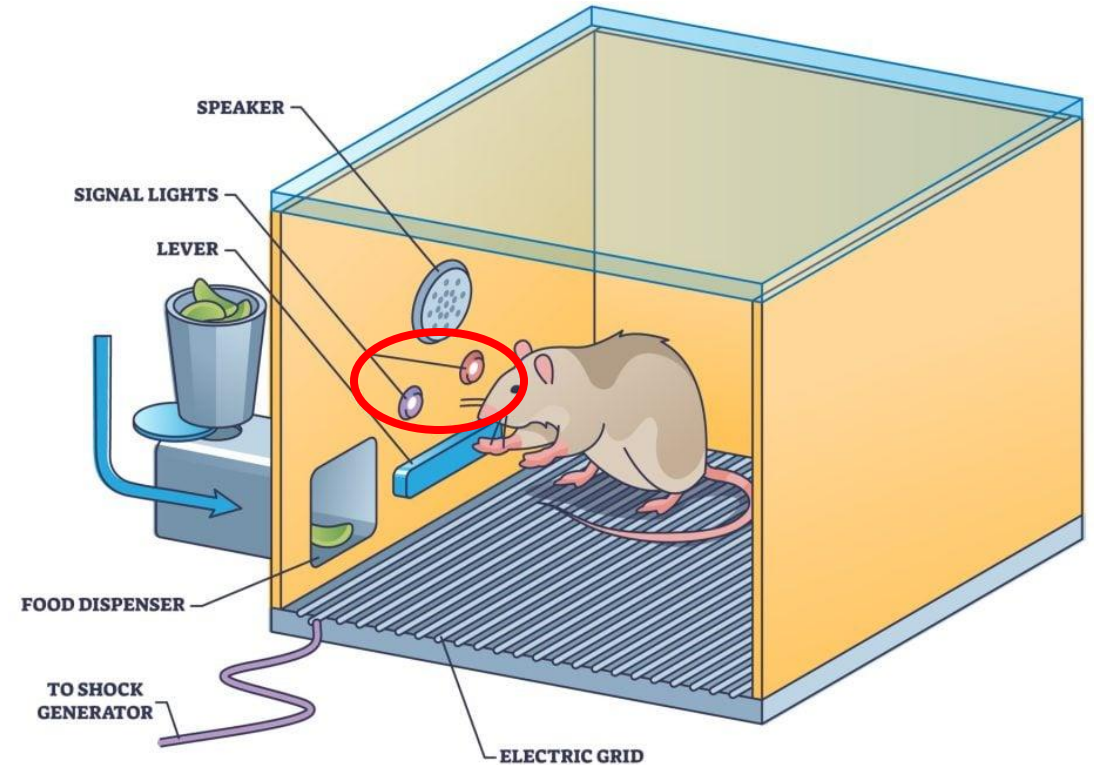
Which is what here

Light : Lever Press → Food

Antecedent

AKA – Discriminative
'Occasion'

Stimulus in the presence of which r
absence of which the



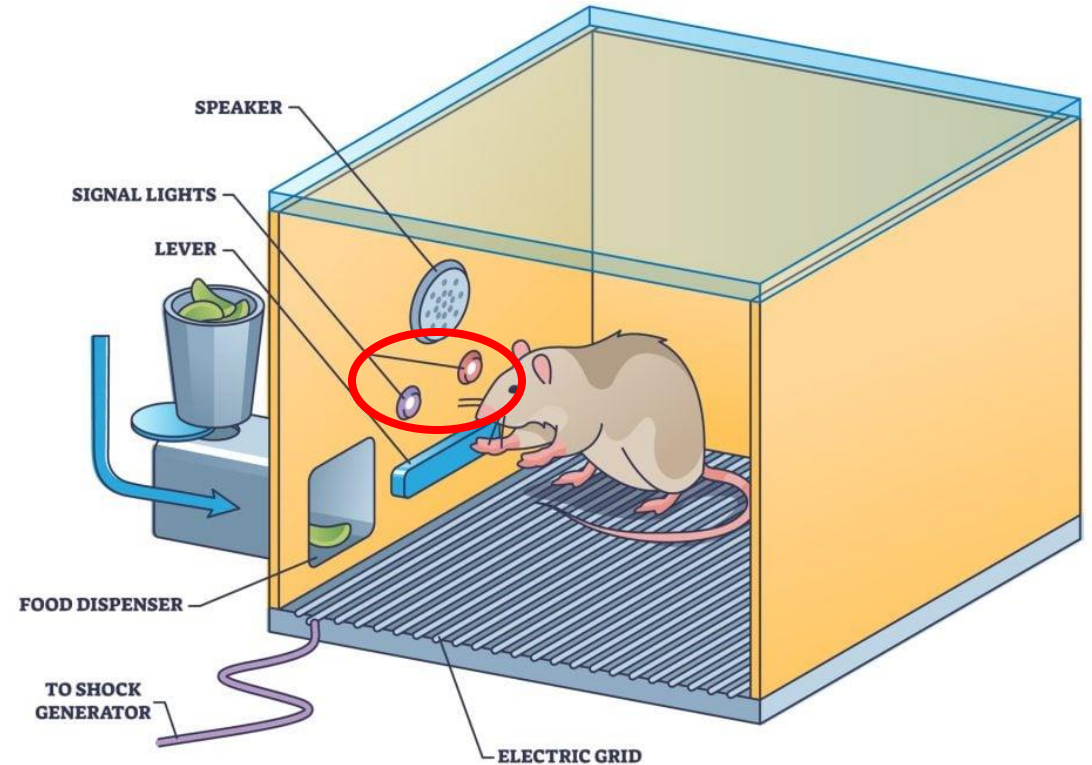
Practical Aspects

Light (S^D): Lever Press (R) $\rightarrow$ Food (S^R)

Antecedent

AKA – Discriminative
'Occasion'

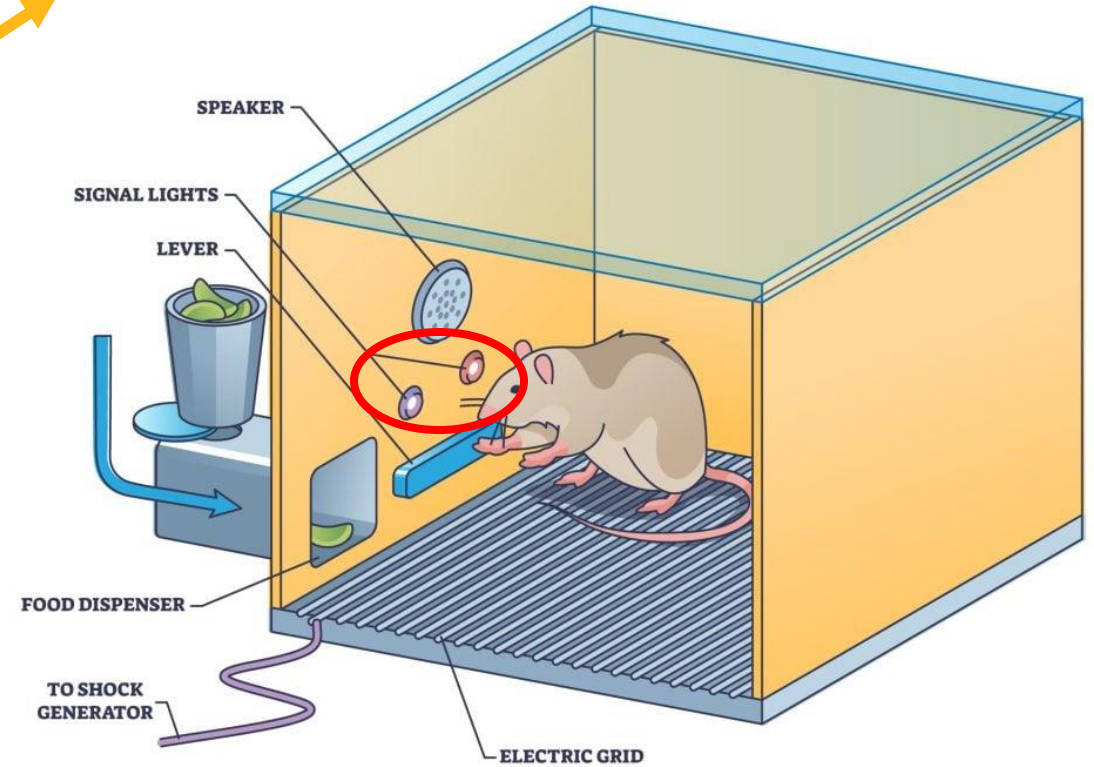
Stimulus in the presence of which r
absence of which the



Practical Aspects

Light (S^D): Lever Press (R) $\rightarrow$ Food (S^R)

Three-Term Contingency



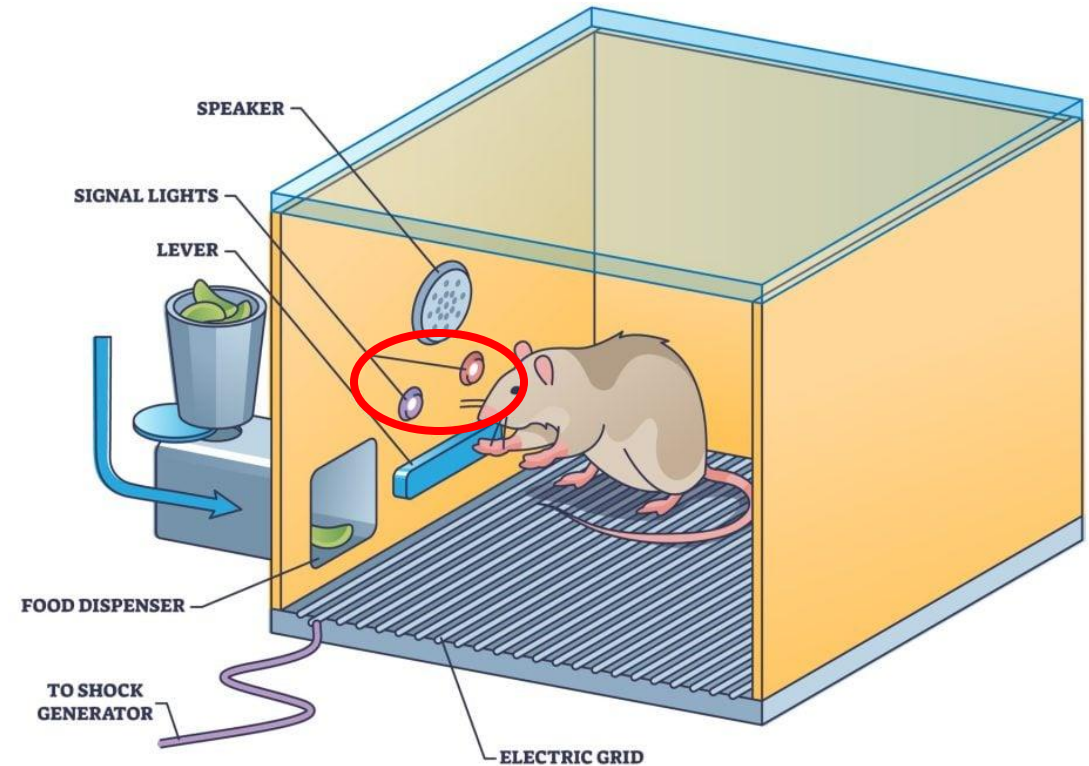
Practical Aspects

Light (S^D): Lever Press (R) $\rightarrow$ Food (S^R)

Operant & Classical – Blurring the Line

Light as S^D associates with food
So, through this paradigm:

Light (CS) $\rightarrow$ Salivate (CR)
Lever Press (R) $\rightarrow$ Food (S^R)



Practical Aspects

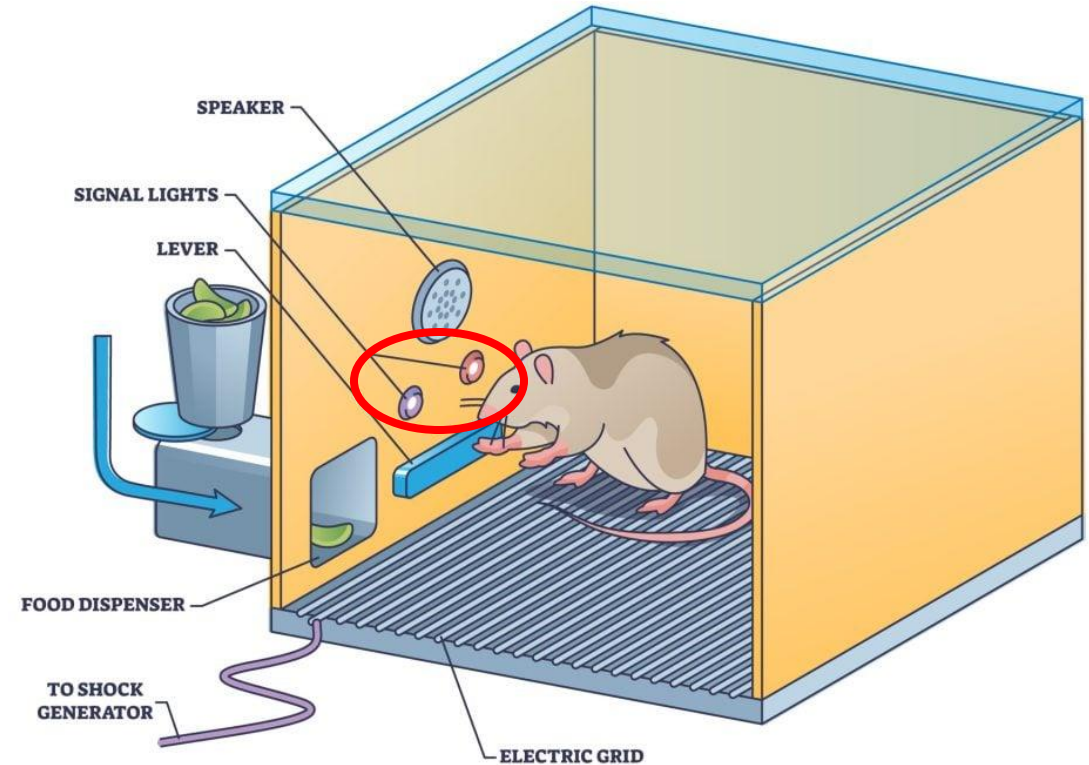
Light (S^Δ): Lever Press (R) $\rightarrow$ Food (—)

Extinction

Weakening of a behavior through the lack of consequence of a previously conseqenced behavior

Discriminative Stimulus for Extinction

Presence of antecedent S signals lack of reinforcing consequence

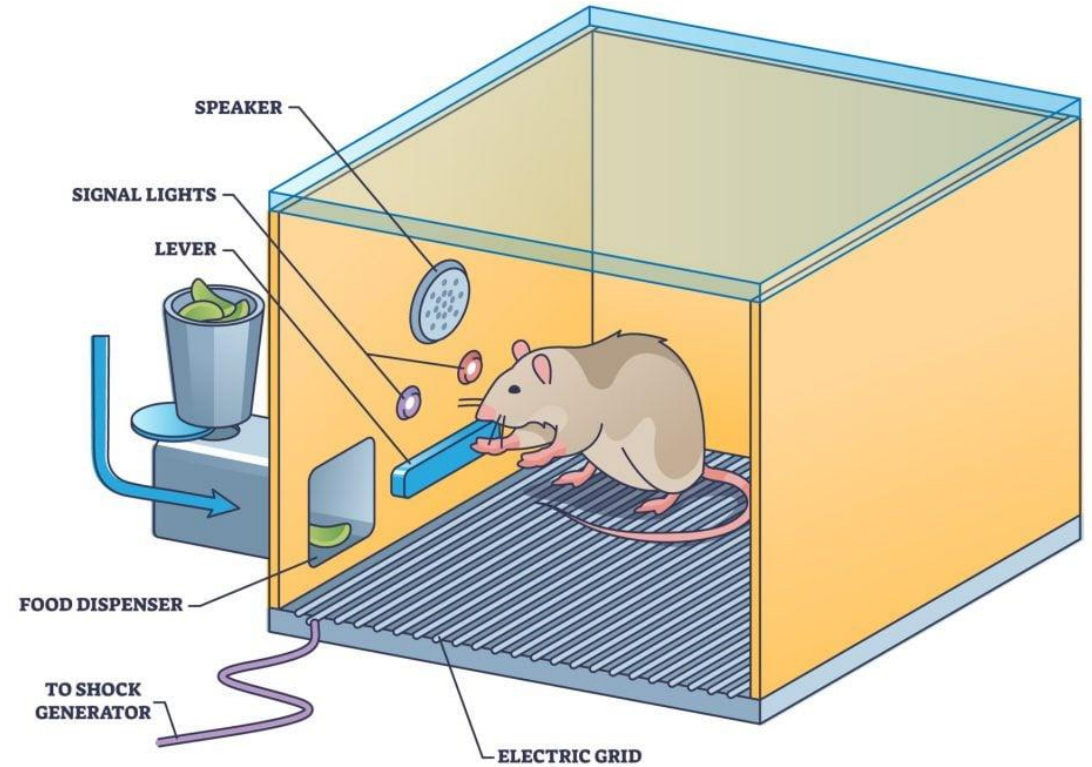




What kind of stimulus
(consequence) is a shock?

Practical Aspects

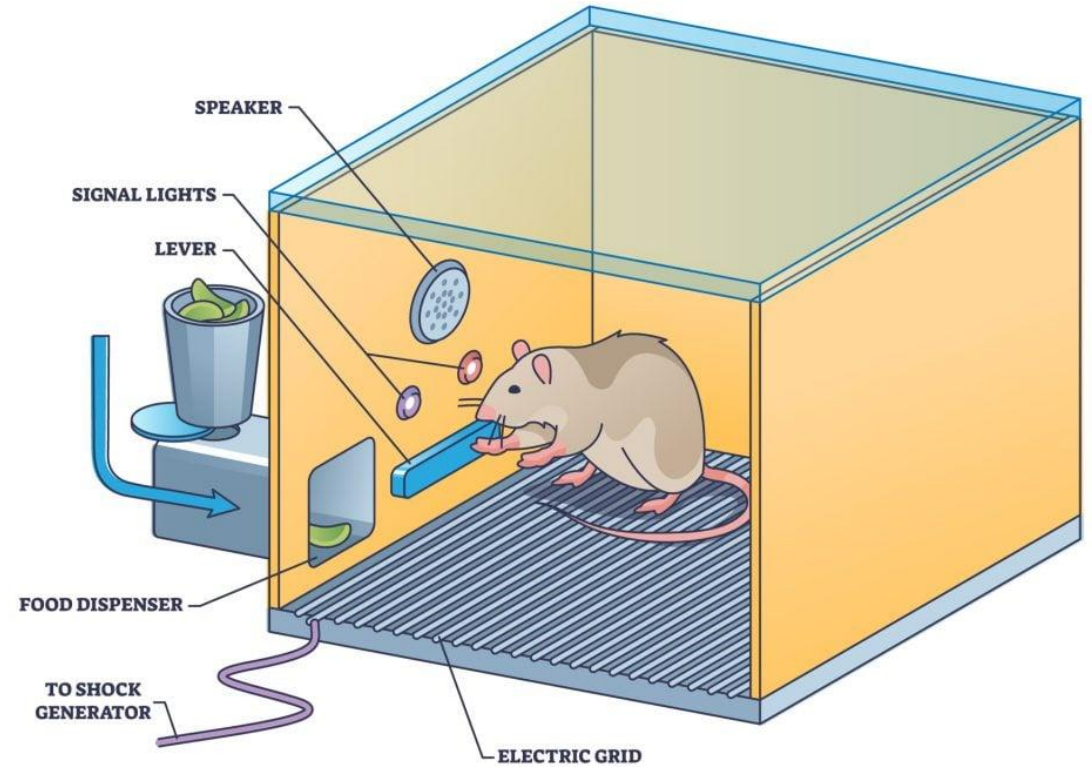
Light (L): Lever Press (R) → Shock (S)



Practical Aspects

Light (L): Lever Press (R) → Shock (S^P)

What if the consequence is a punisher?



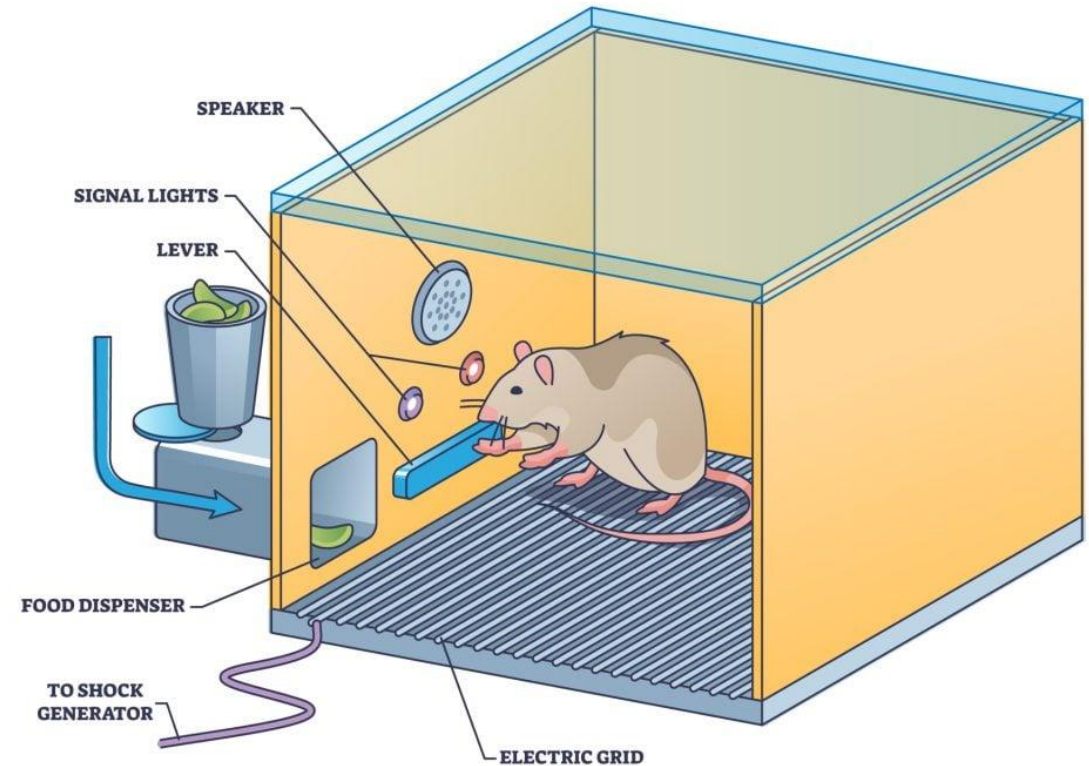
Practical Aspects

Light (S^{Dp}): Lever Press (R) → Shock (S^P)

What if the consequence is a punisher?

The light becomes a **discriminative stimulus for punishment**

*An antecedent that signals the response **will** be punished*



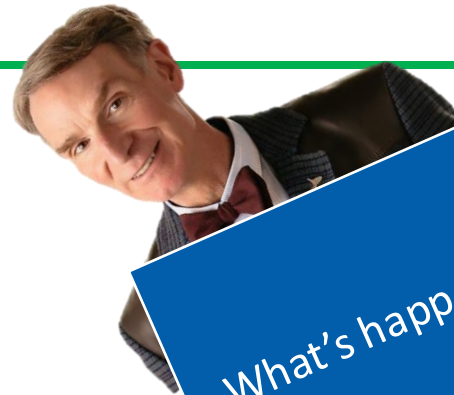
Practical Aspects

What if the consequence is a punisher?

The light becomes a **discriminative stimulus for punishment**

An antecedent that signals the response will be punished

Cop (S^{Dp}): Speed (R) $\rightarrow$ Ticket (S^P)



What's happening here?

Practical Aspects

What if the consequence is a punisher?

The light becomes a **discriminative stimulus for punishment**

An antecedent that signals the response will be punished

Cop (S^{Dp}): Speed (R) $\rightarrow$ Ticket (S^P)

Cop is punishment for speeding, learn to not speed when you see a cop because you will get a ticket

Practical Aspects

What if the consequence is a punisher?

The light becomes a **discriminative stimulus for punishment**

An antecedent that signals the response will be punished

For your dog:

Spray Bottle (S^{Dp}): Bark (R) $\rightarrow$ Sprayed (S^P)



And here?

Practical Aspects

What if the consequence is a punisher?

The light becomes a **discriminative stimulus for punishment**

An antecedent that signals the response will be punished

For your dog:

Bottle (S^{Dp}): Bark (R) $\rightarrow$ Sprayed (S^P)

They learn that, if the spray bottle comes out, a bark will be punished – no barking when bottle is out

Practical Aspects



Recap...?

Types of Antecedents

Discriminatory Stimulus

Discriminatory Stimulus for Punishment

Discriminative Stimulus for Extinction

Practical Aspects

Discriminative Stimulus [for Reinforcement]

(S^D)

An antecedent that signals the response will be reinforced

Discriminative Stimulus for Punishment

(S^{Dp})

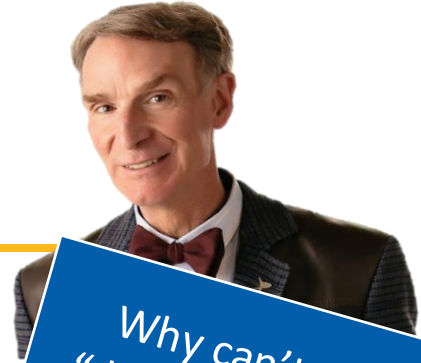
An antecedent that signals the response will be punished

Discriminative Stimulus for Extinction

(Extinction Stimulus, S^A)

An antecedent that signals the response will not be reinforced

Practical Aspects



Discriminative Stimulus

(S^D)

An antecedent that signals the response will be reinforced

Discriminative Stimulus for Punishment

(S^{Dp})

An antecedent that signals the response will be punished

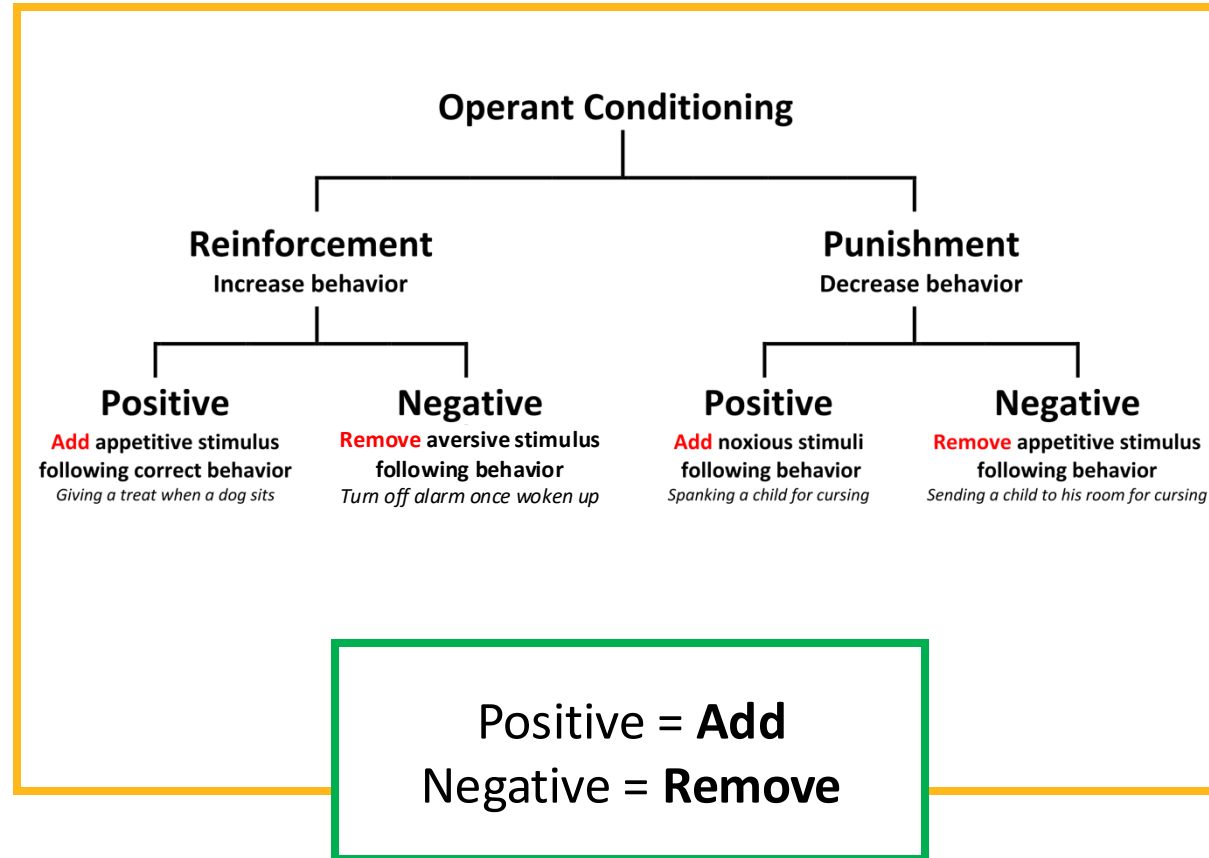
Discriminative Stimulus for Extinction

(Extinction Stimulus, S^A)

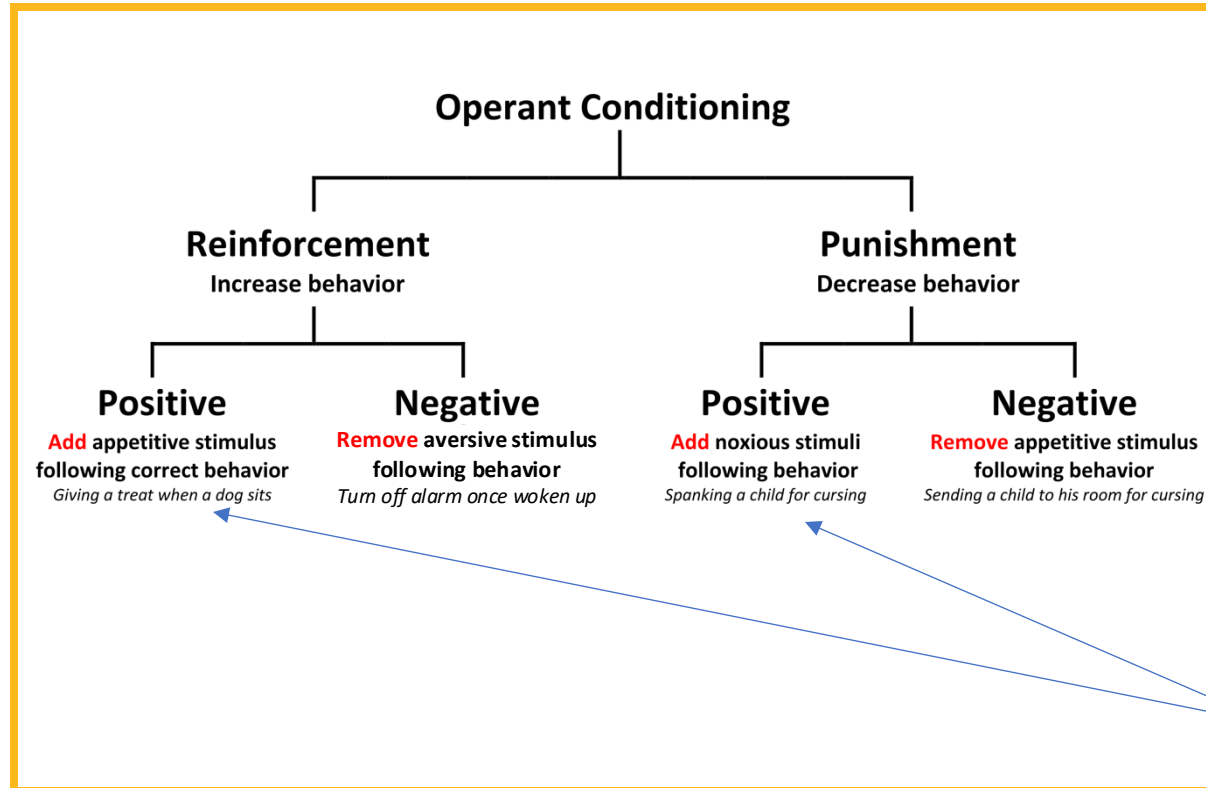
An antecedent that signals the response will not be reinforced

Why can't we have a
"discriminative stimulus
for extinction of
punishment"?

Contingencies

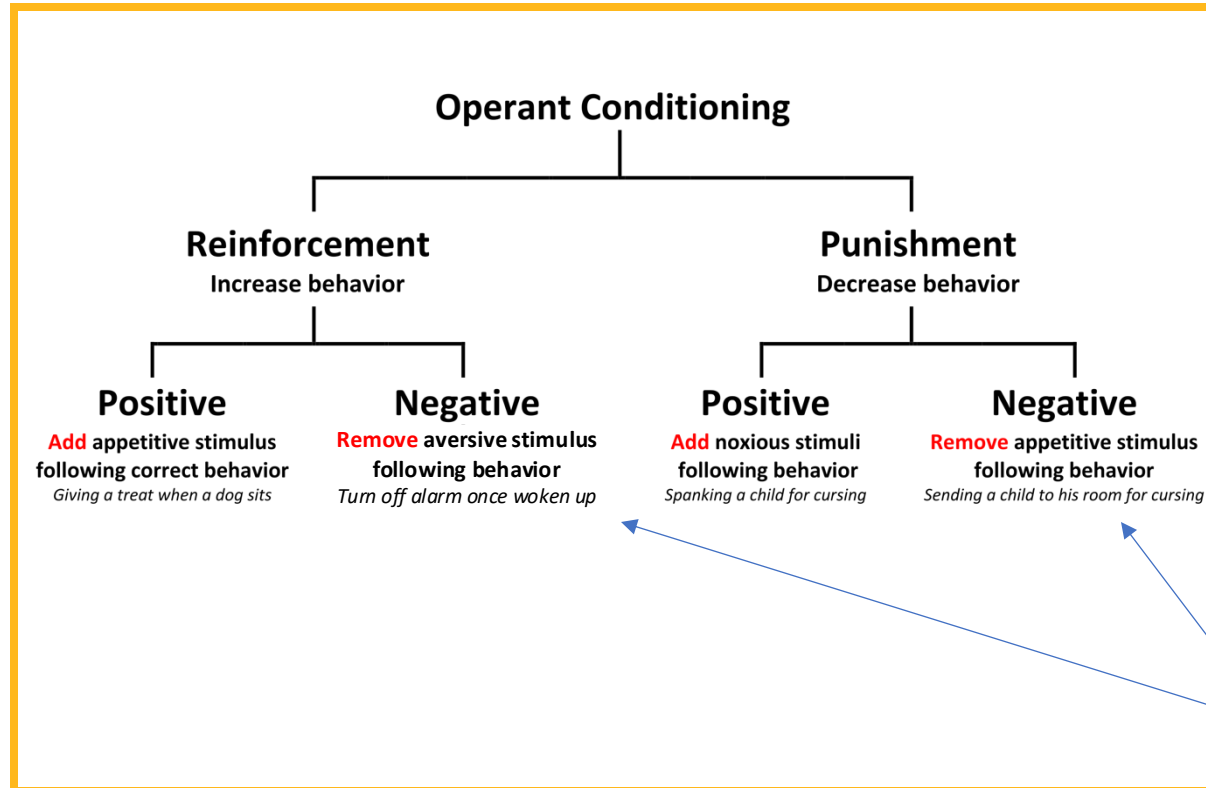


Contingencies



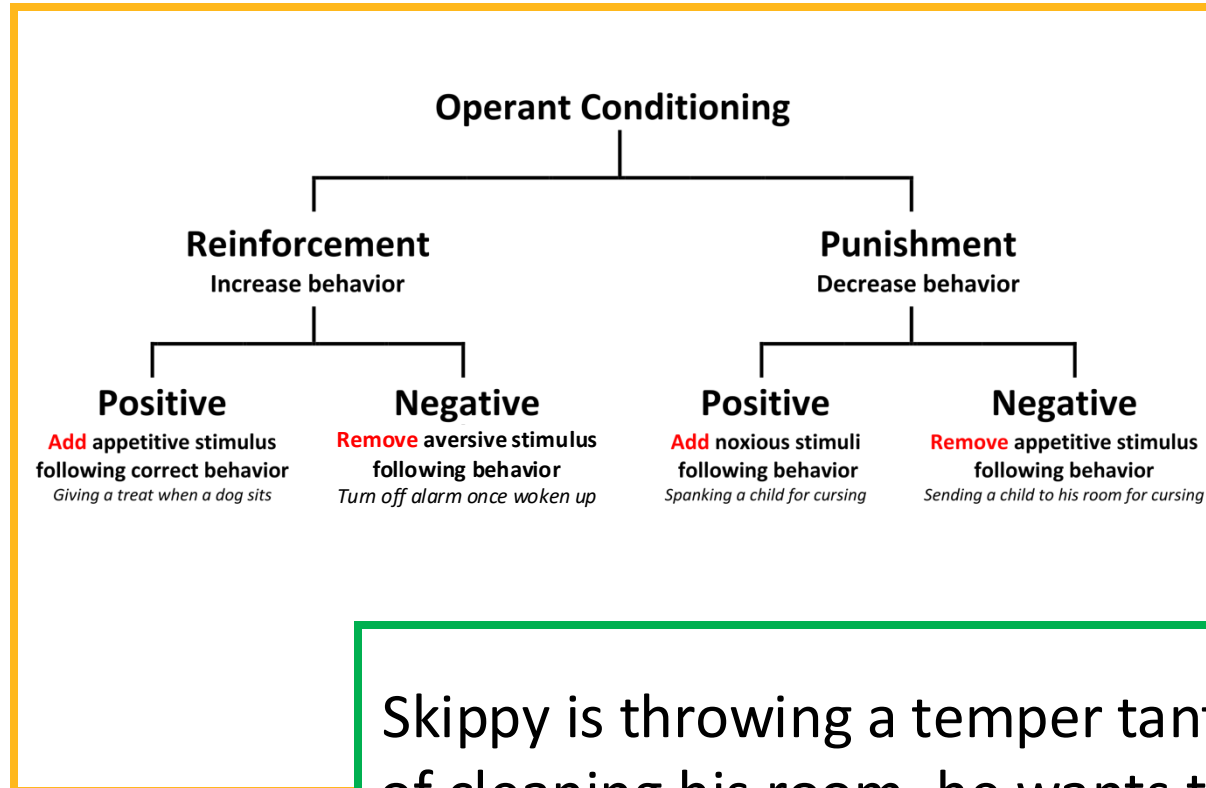
Can we think of some examples of these two?

Contingencies



What about these two...?

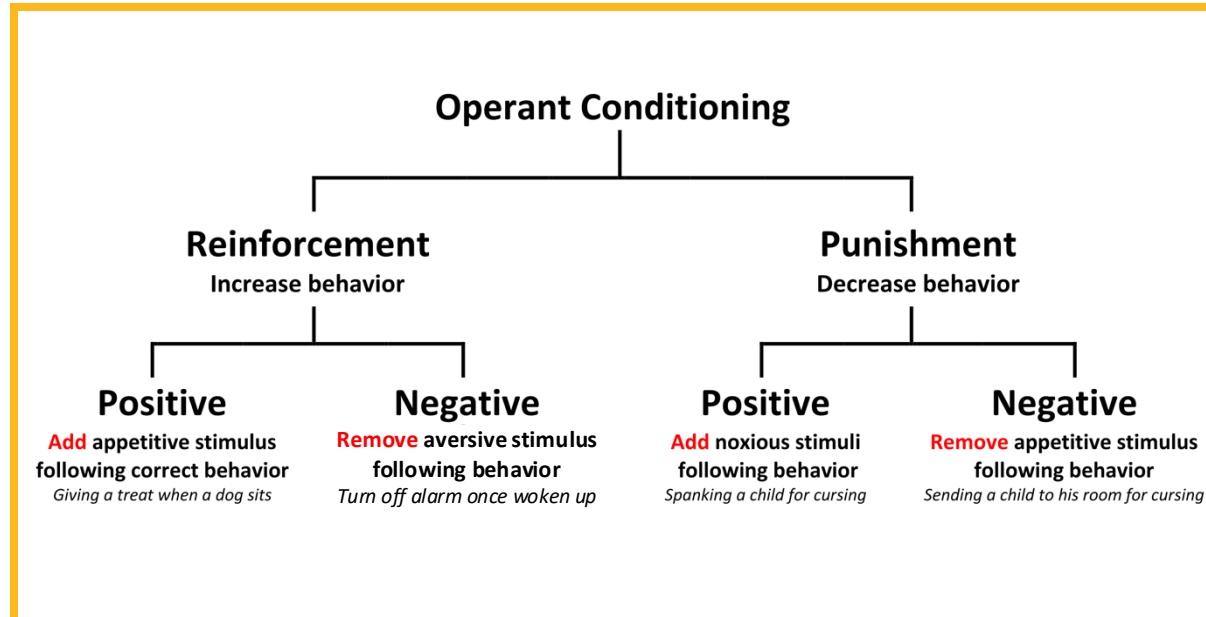
Contingencies



What's this an example of?

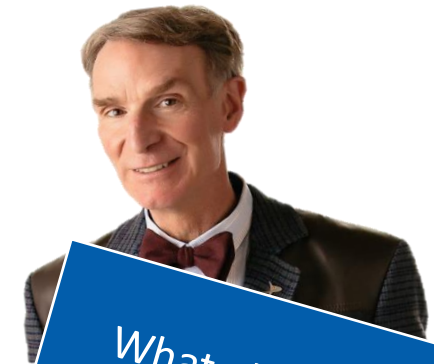
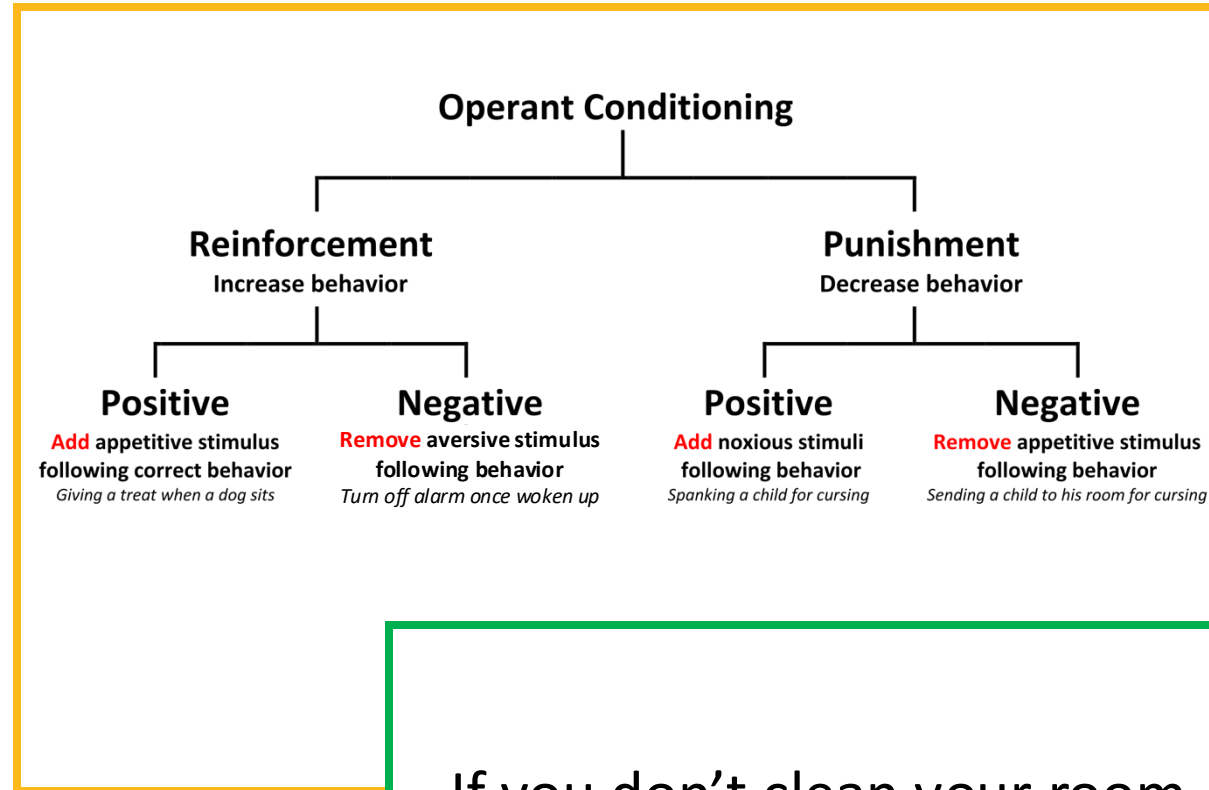
Skippy is throwing a temper tantrum because he's tired of cleaning his room, he wants to play Gameboy. So, his mom gives him his Gameboy.

Contingencies



Skippy is temper tantrum-ing → Mom gives Gameboy
Positive Reinforcement

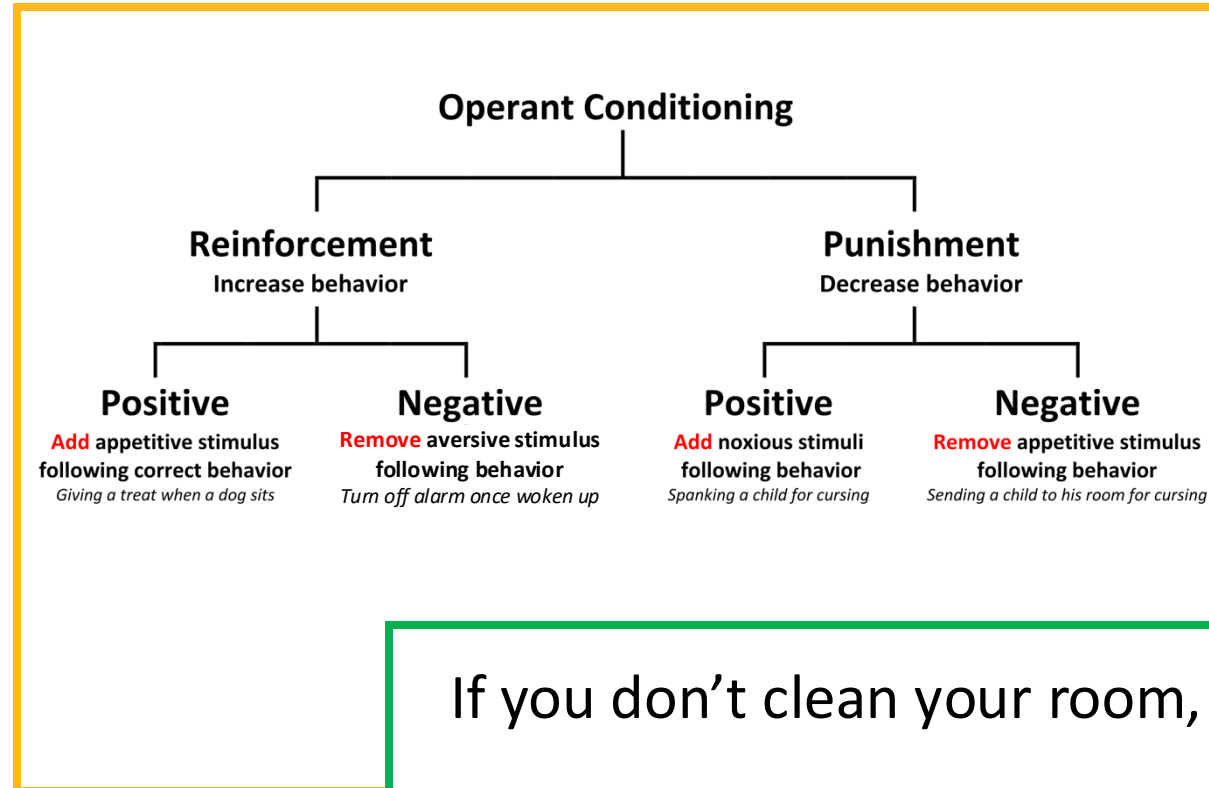
Contingencies



What about this one?

If you don't clean your room, you won't get dessert

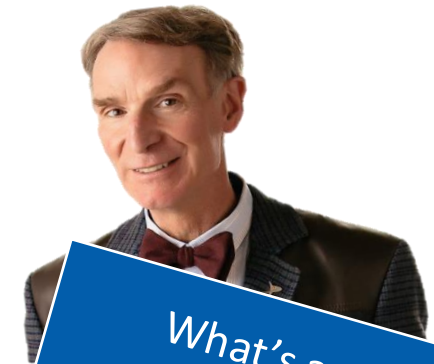
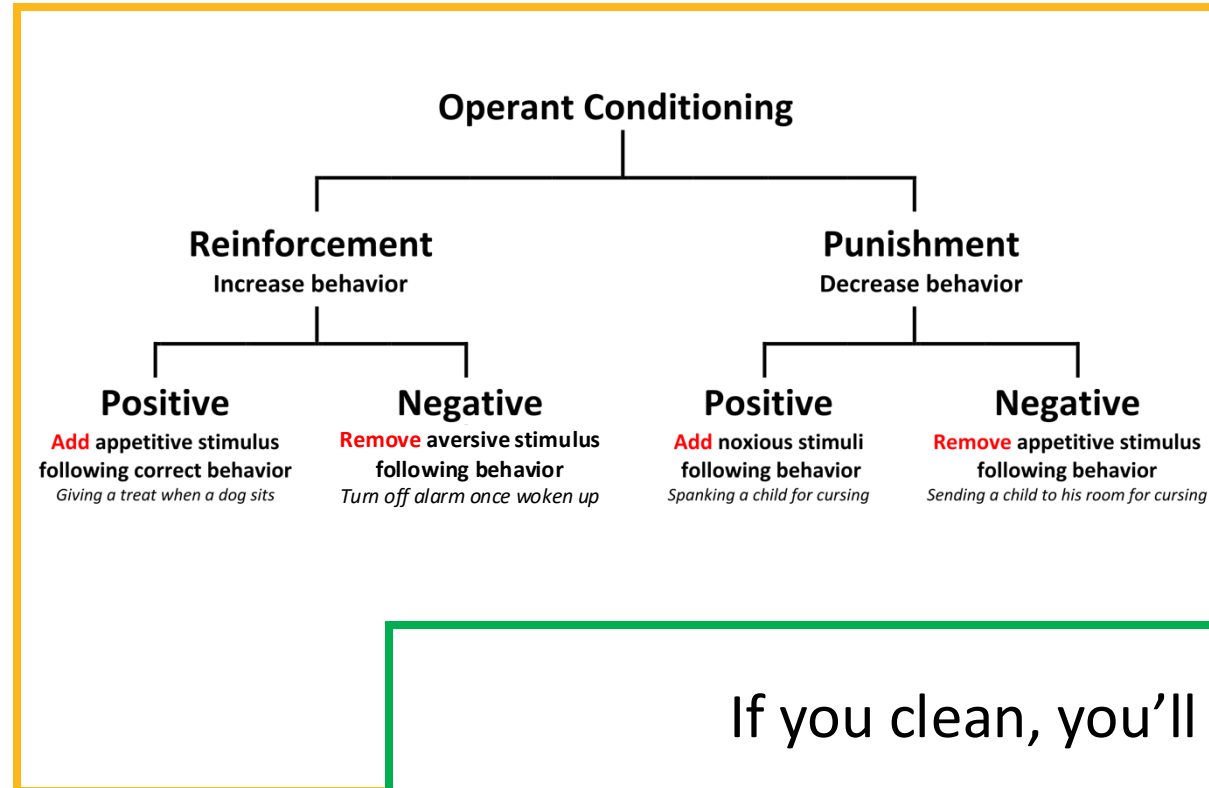
Contingencies



If you don't clean your room, you won't get dessert

*Can't reinforce/punish the lack of a behavior
Re-word paradigm to be about a behavior, not the lack of one*

Contingencies

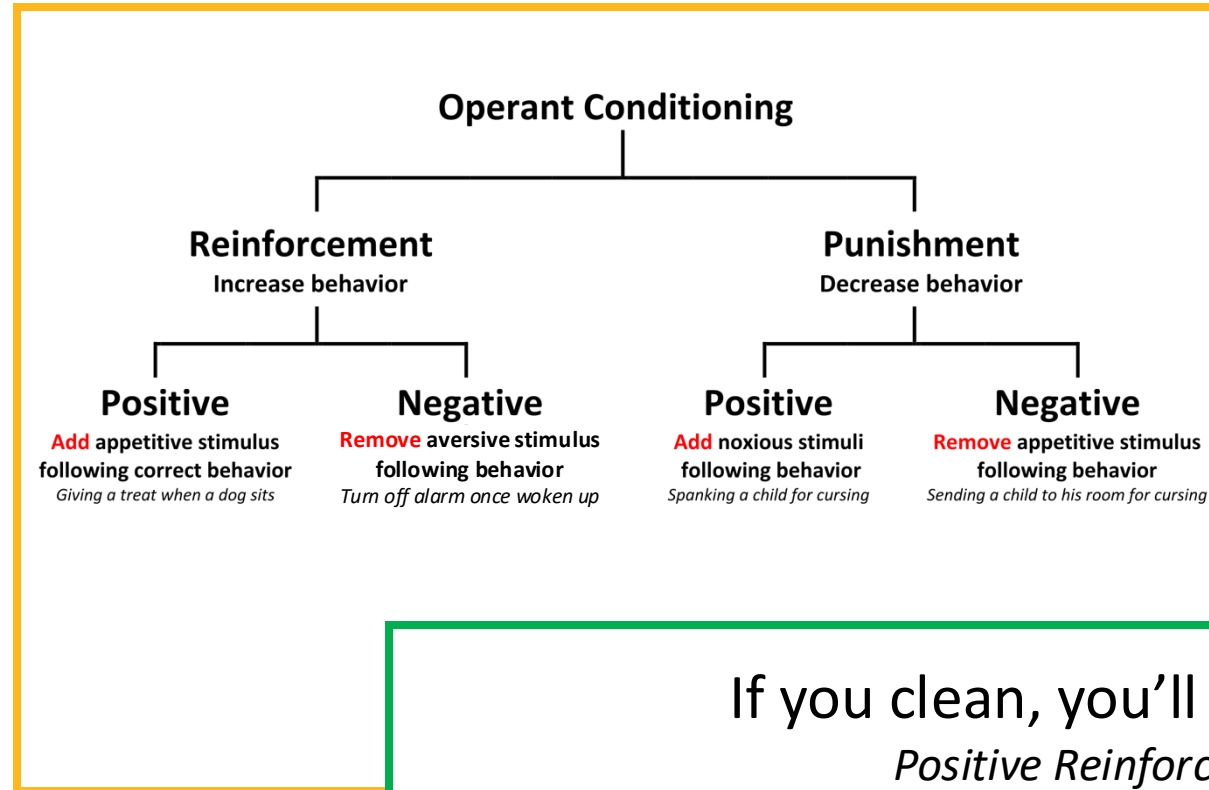


What's are these examples of?

If you clean, you'll get dessert

If you game, you won't get dessert

Contingencies



If you clean, you'll get dessert

Positive Reinforcement

If you game, you won't get dessert

Negative Punishment

Contingencies

Negative Reinforcement

Remove an aversive stimulus

Negative Punishment

Remove an appetitive stimulus

Positive Reinforcement

Add an appetitive stimulus

Positive Punishment

Add an aversive Stimulus

Contingencies

Lever Press (R) → Remove Shock (S^R)

Negative Reinforcement
Remove an aversive stimulus

Lever Press (R) → Receive Shock (S^P)

Negative Punishment
Remove an appetitive stimulus

Lever Press (R) → Receive Food (S^R)

Positive Reinforcement
Add an appetitive stimulus

Lever Press (R) → Remove Food (S^P)

Positive Punishment
Add an aversive Stimulus

Types of Stimuli

(consequences)

Immediate vs Delayed

Primary vs Secondary

Intrinsic vs Extrinsic

Natural vs Contrived

Types of Stimuli

- **Immediate vs Delayed**
 - The quicker the S^R / S^P , the stronger its effect
 - In humans, loss of effect over time can be assuaged by communicating or having expectations of a delayed S^R / S^P
- Primary vs Secondary
- Intrinsic vs Extrinsic
- Natural vs Contrived

Types of Stimuli:

Finding out you got an "A" day-of is better than 1 week later
Expectation that you won't find out for a week can diminish negative effect of delay

- **Immediate vs Delayed**

- The quicker the S^R / S^P , the stronger its effect
- In humans, loss of effect over time can be assuaged by communicating or having expectations of a delayed S^R / S^P
- Primary vs Secondary
- Intrinsic vs Extrinsic
- Natural vs Contrived

If dealing with an organism without communication (dog, baby), immediate reinforcement is **HUGELY** important

Types of Stimuli

- Immediate vs Delayed
- **Primary vs Secondary**
 - Primary S^R / S^P – innate
 - Secondary S^R / S^P – learned
- Intrinsic vs Extrinsic
- Natural vs Contrived

Types of Stimuli

- Immediate vs Delayed
- **Primary vs Secondary**
 - Primary (Unconditioned) S^R / S^P – innate
 - Secondary (Conditioned) S^R / S^P – learned
- Intrinsic vs Extrinsic
- Natural vs Contrived

Types of Stimuli:

Lever Press (R) $\rightarrow$ Food (S^R)

- Immediate vs Delayed
- **Primary vs Secondary**
 - **Primary** (Unconditioned) S^R / S^P – innate
 - **Secondary** (Conditioned) S^R / S^P – learned
- Intrinsic vs Extrinsic
- Natural vs Contrived

Types of Stimuli:

Tone (CS) : Food (US) $\rightarrow$ Salivate
Lever Press (R) $\rightarrow$ Tone (S^R)

- Immediate vs Delayed
- **Primary vs Secondary**
 - Primary (Unconditioned) S^R/ S^P– innate
 - **Secondary** (Conditioned) S^R/ S^P – learned
- Intrinsic vs Extrinsic
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Types of Stimuli

- Immediate vs Delayed
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 - Primary (Unconditioned) S^R / S^P – innate
 - **Secondary** (Conditioned) S^R / S^P – learned
- Intrinsic vs Extrinsic
- Natural vs Contrived



Money...?

Types of Stimuli

- Immediate vs Delayed
- **Primary vs Secondary**
 - Primary (Unconditioned) S^R / S^P – innate
 - Secondary (Conditioned) S^R / S^P – learned
 - Learned Industriousness Theory – ‘hard work’ can become a secondary reinforcer when it has been rewarded in the past with S^R / S^P
- Intrinsic vs Extrinsic
- Natural vs Contrived

Types of Stimuli

- Immediate vs Delayed
- **Primary vs Secondary**
 - Primary (Unconditioned) S^R / S^P – innate
 - Secondary (Conditioned) S^R / S^P – learned
 - Learned Industriousness Theory – rats that have been rewarded for **stronger** lever presses (rewarded for ‘harder’ work) will run **faster** down an alley to get a reward
- Intrinsic vs Extrinsic
- Natural vs Contrived

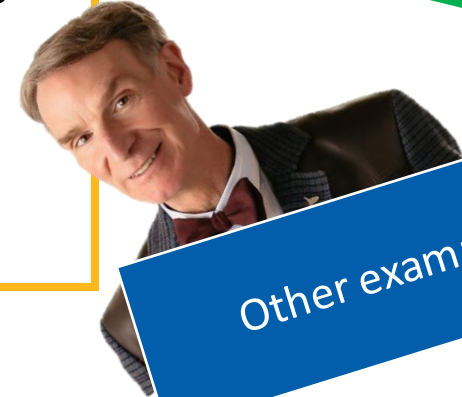
Types of Stimuli

- Immediate vs Delayed
- Primary vs Secondary
- **Intrinsic vs Extrinsic**
 - Intrinsic S^R / S^P – R itself *is* S^R / S^P
 - Extrinsic S^R / S^P – external consequence is S^R / S^P
- Natural vs Contrived

Types of Stimuli

- Immediate vs Delayed
- Primary vs Secondary
- **Intrinsic vs Extrinsic**
 - **Intrinsic** S^R / S^P – R itself is S^R / S^P
 - Extrinsic S^R / S^P – external consequence is S^R / S^P
- Natural vs Contrived

Socialize because we enjoy our friends' company
Play soccer because we love the sport



Other examples??

Types of Stimuli

- Immediate vs Delayed
- Primary vs Secondary
- **Intrinsic vs Extrinsic**
 - Intrinsic S^R / S^P – behavior (R) itself is S^R / S^P
 - **Extrinsic** S^R / S^P – external consequence is S^R / S^P
- Natural vs Contrived

Socialize because it will make our partner happy
Play soccer because it's for a scholarship

Types of Stimuli

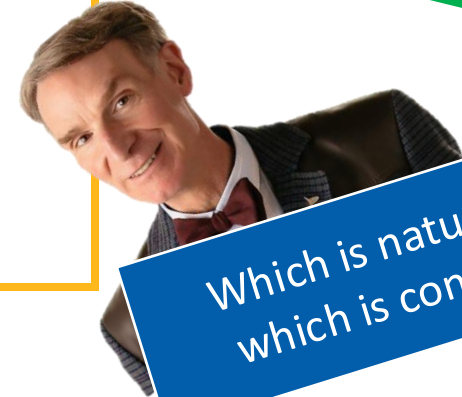
- Immediate vs Delayed
- Primary vs Secondary
- Intrinsic vs Extrinsic
- **Natural vs Contrived**
 - Natural S^R / S^P – typical, expected consequence
 - Contrived S^R / S^P – arranged consequence

Types of Stimuli

- Immediate vs Delayed
- Primary vs Secondary
- Intrinsic vs Extrinsic
- **Natural vs Contrived**
 - Natural S^R / S^P – typical, expected consequence
 - Contrived S^R / S^P – arranged consequence

Socialize to reduce arguing with partner
vs.

Socialize to feel happy



*Which is natural and
which is contrived?*

Questions?